ABSOLUTE VALUES

Def'n

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$$

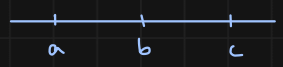
Theorem

<u>Triangle Inequality</u>

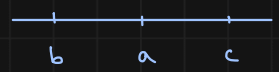
For any $a, b, c \in \mathbb{R}$

$$|a - c| \leq |a - b| + |b - c|$$

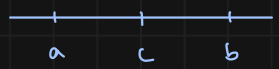
<u>Proof</u>:

Case 1: 

$|a - c| = |a - b| + |a - c|$

Case 2: 

$|a - c| < |b - c| + |a - b|$

Case 3: 

$|a - c| < |a - b| + |b - c|$

Corollary

<u>Triangle Inequality pt. 2</u>

$$|x + y| \leq |x| + |y|$$

<u>Proof</u>: Let $a = x$, $b = 0$, $c = -y$

$$|a - c| \leq |a - b| + |b - c|$$
$$|x - (-y)| \leq |x - 0| + |0 - (-y)|$$
$$|x + y| \leq |x| + |y|$$

Def'n

$$|f(x)| = \begin{cases} f(x) & \text{when } f(x) \geq 0 \\ -f(x) & \text{when } f(x) < 0 \end{cases}$$

QUIZ #1, PRACTICE PROBLEMS #1

PRACTICE PROBLEMS

1. (a) $|x - 2| = 5$

Case 1: $x < 0$

$$-(x-2) = 5$$
$$-x + 2 = 5$$
$$x = -3$$

Case 2. $x \geq 0$

$$x - 2 = 5$$
$$x = 7$$

(b) $|x - 4| = |3x + 2|$

Case 1: $x < -\frac{2}{3}$

$$-(x-4) = -(3x+2)$$
$$-x + 4 = -3x - 2$$
$$2x = -6$$
$$x = -3$$

Case 2: $-\frac{2}{3} \leq x \leq 4$

$$-(x-4) = 3x + 2$$
$$-x + 4 = 3x + 2$$
$$-4x = -2$$
$$x = \frac{1}{2}$$

Case 3: $x > 4$

$$x - 4 = 3x + 2$$
$$-2x = 6$$
$$x = -3$$

(c) $1 < |x + 1|$

Case 1 $x < -1$

$$1 < -(x+1)$$
$$1 < -x - 1$$
$$-2 > x$$

Case 2: $x \geq -1$

$$1 < x + 1$$
$$0 < x$$

(d) $|x + 3| + |1 - 2x| \leq 5$

Case 1: $x < -3$

$$-(x+3) + 1 - 2x \leq 5$$
$$-x - 3 + 1 - 2x \leq 5$$
$$-3x \leq 7$$
$$x \geq -\frac{7}{3} \Rightarrow x \; 4 \; -3$$

Case 2: $-3 \leq x \leq -\frac{1}{2}$

$$x + 3 + 1 - 2x \leq 5$$
$$-x \leq 1$$
$$x \geq 1$$

Case 3: $x > -\frac{1}{2}$

$$x + 3 - (1 - 2x) \leq 5$$
$$x + 3 - 1 + 2x \leq 5$$
$$3x \leq 3$$
$$x \leq 1$$

(e) $|x-4||x+2| > 7$

Case 1. $x < -2$

$$-(x-4) \cdot -(x+2) > 7$$
$$(-x+4)(-x-2) > 7$$
$$x^2 - 2x - 15 > 0 \Rightarrow x = -3, \quad x \neq 5 < -2$$

Case 2: $-2 \leq x \leq 4$

$$-(x-4)(x+2) > 7$$
$$(-x+4)(x+2) > 7$$
$$-x^2 + 2x + 1 > 0 \Rightarrow 1 - \sqrt{2} < x < 1 + \sqrt{2}$$

Case 3. $x > 4$

$$(x-4)(x+2) > 7$$
$$x^2 - 2x - 15 > 0 \Rightarrow x = 5, \quad x \neq -3 > 4$$

2. (a) Let $x = a$, $y = -b$, $z = 0$

$$|x-z| \leq |x-y| + |y-z|$$
$$|a-0| \leq |a-(-b)| + |-b|$$
$$|a| - |b| \leq |a+b|$$

(b) Let $x = a$, $y = b$, $z = 0$

$$|x-z| \leq |x-y| + |y-z|$$
$$|a-0| \leq |a-b| + |b-0|$$
$$|a| - |b| \leq |a-b|$$

3. $\frac{1}{2}(a+b+|a-b|)$ calculates the max / min

Case 1. $a \geq b$

$$\frac{1}{2}(a+b+a-b) = \frac{1}{2}(2a)$$
$$= a$$

Case 2. $a \leq b$

$$\frac{1}{2}(a+b-(a-b)) = \frac{1}{2}(a+b-a+b)$$
$$= \frac{1}{2}(2b)$$
$$= b$$

$$\frac{1}{2}(a+b+|a-b|) = \begin{cases} a, & \text{if } a > b \\ b, & \text{if } a < b \end{cases}$$

4. (a) F
(b) T
(c) F
(d) F
(e) T

5. (a)
(b)
(c)
(d)
(e)
(f)

Case 1: $x < 1$

$$\pi < -x + 1 \leq 5$$
$$\pi - 1 < -x \leq 4$$
$$-\pi + 1 > x \geq -4$$

Case 2: $x > 1$

$$\pi < x - 1 \leq 5$$
$$\pi + 1 < x \leq 6$$

$\therefore [-4, -\pi + 1) \cup (\pi + 1, 6]$

$\therefore$ (c), (d), (f)